

Equipping Llama with Google Query API for Improved Accuracy and Reduced Hallucination

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Equipping Llama with Google Query API for Improved Accuracy and Reduced Hallucination

Yeong Sik Bae, Hye Rin Kim, and Jae Hoon Kim

Abstract

This study investigates the integration of the Llama 2 7b large language model (LLM) with the Google Query API to enhance its accuracy and reduce hallucination instances. By leveraging real-time internet data, we aimed to address the limitations of static training datasets and improve the model's performance across various language processing tasks. The methodology involved augmenting Llama 2 7b's architecture to incorporate dynamic data retrieval from the Google Query API, followed by an evaluation of its impact on model accuracy and hallucination reduction using the BIG-Bench benchmark. The results indicate significant improvements in both accuracy and reliability, demonstrating the effectiveness of integrating LLMs with external data sources. This integration not only marks a substantial advancement in the capabilities of LLMs but also raises important considerations regarding data bias, privacy, and the ethical use of internet-sourced information. The study's findings contribute to the ongoing discourse on enhancing LLMs, suggesting a promising direction for future research and development in artificial intelligence.

Keywords: Artificial Intelligence, Data Integration, Google Query API, Hallucination Reduction, Language Models, Llama 2 7b, Model Accuracy

1. Introduction

Large language models (LLMs) have revolutionized the field of natural language processing (NLP), offering unprecedented capabilities in generating human-like text, understanding context, and answering questions across various domains [1, 2]. Among these models, Llama 2 7b, developed by Meta Inc., stands out due to its remarkable efficiency and accuracy, powered by 7 billion parameters. Such models have been instrumental in pushing the boundaries of what LLMs can achieve, providing researchers and developers with a versatile tool for a wide range of applications. However, despite its strengths, Llama 2 7b, like other LLMs, is not immune to the challenges that plague these advanced systems, particularly the issue of generating text that, while plausible, may not be grounded in factual accuracy, a phenomenon known as "hallucination" [2, 3].

Hallucination in LLMs poses significant challenges, especially when those models are used in scenarios requiring high levels of trust and accuracy, such as in news generation, educational content, and sensitive decision-making contexts [3]. The phenomenon not only undermines the reliability of the models but also raises concerns about their applicability in critical applications [3]. Hallucination is attributed to various factors, including the model's reliance on patterns in the training data rather than factual understanding or real-world knowledge, and the assessors' own subjective understanding of the LLM evaluation criteria [3]. Addressing this issue is crucial for advancing the utility and reliability of LLMs in producing content that is both engaging and accurate.

The proposed solution to mitigate hallucination involves integrating Llama 2 7b with the Google Query API, leveraging

the vast repository of Internet-available content to verify facts and ensure the generated text is anchored in reality. This integration aims to enhance the model's accuracy by providing it with access to real-time information, enabling a more informed and precise text generation process. By querying the Google API, Llama 2 7b can cross-reference information before including it in the generated text, significantly reducing instances of hallucination. This approach not only promises to improve the reliability of Llama 2 7b but also sets a precedent for the development of more robust and trustworthy LLMs. The following sections will delve into the background of Llama 2 7b, explore the integration methodology, and present an analysis of the impact of this solution on reducing hallucination and improving overall model accuracy.

The rest of the article structure is described as below: Section 2 is the background. Section 3 is the methodology. Section 4 is the implementation details. Section 5 is the experiment results. Section 6 is the discussion. Section 7 is the conclusion.

2. Background

This section reviews the existing literature on efforts to enhance the accuracy and reduce the hallucinations of LLMs, structured around four key themes that have emerged as critical in the discourse on language model development and optimization.

2.1. Enhancing LLM Accuracy

Studies have consistently shown that the accuracy of language models can be significantly improved through advanced

training techniques, including transfer learning and unsupervised pre-training, which enable models to leverage vast datasets more effectively [4, 5, 6, 7, 8, 9]. Research has also highlighted the importance of dynamic data refreshing, where continuously updating the training corpus with recent information can help maintain the model’s relevance over time [10, 11, 12]. Moreover, incorporating external knowledge bases has been found to enhance the model’s ability to generate factually accurate responses [13, 14, 15, 16]. The use of multimodal data, integrating text with images or videos, has further been shown to improve the contextual understanding of those models [17, 18, 19]. Techniques for optimizing model architecture, such as attention mechanisms and transformer layers, have been pivotal in increasing the efficiency and accuracy of LLMs [20, 8, 21, 22]. Adaptive learning rates and fine-tuning strategies have also been identified as crucial for tailoring models to specific tasks or domains, thereby improving their performance [8, 14, 23, 24]. Lastly, the implementation of robust evaluation metrics enables a more nuanced assessment of model accuracy, guiding iterative improvements [17, 25, 14, 26, 16].

2.2. Mitigating Hallucination in LLMs

Efforts to reduce hallucination in language models have focused on several key strategies, including the development of more sophisticated training algorithms that can better distinguish between factual and fabricated content [27, 28, 14, 29, 30]. The integration of verification mechanisms, which cross-reference generated content with trusted data sources, has been shown to curb the incidence of hallucinations significantly [27, 3]. Implementing feedback loops where models are regularly updated based on their performance in real-world applications helps in identifying and correcting patterns leading to hallucinations [31, 32, 33, 27, 34]. The use of constrained generation techniques, which limit the model’s ability to generate content outside certain factual boundaries, has also been effective [34, 35, 36, 37]. Enhancing the models’ understanding of context and causality has led to more coherent and accurate text generation [27, 38, 39]. Research has pointed towards the utility of ensemble methods, combining the outputs of multiple models to filter out hallucinated content [40, 41, 42]. Furthermore, developing specialized training datasets focused on factual accuracy has been instrumental in training models to prioritize verifiable information [43, 27, 44].

2.3. Integrating External APIs for Enhanced Model Performance

The integration of external APIs with language models has opened new avenues for enhancing their capabilities, allowing real-time access to updated information and thus significantly improving the relevance and accuracy of generated content [45, 46, 47, 48]. Such integrations also enable models to perform more complex tasks, such as real-time translations, fact-checking, and information retrieval, directly within the generation process [49, 50]. The use of APIs for dynamic data sourcing helps models stay current with the latest events and information, addressing one of the major challenges in maintaining the utility of LLMs [8, 4, 51]. Leveraging APIs for

specialized tasks, such as sentiment analysis or entity recognition, has been shown to enhance the models’ analytical capabilities [8, 52]. The customization of API integrations based on the specific needs of a task or domain allows for a more targeted approach to model enhancement [24, 53, 54, 55]. The establishment of secure and efficient communication protocols between language models and APIs is crucial for maximizing the benefits of this integration [2, 56, 57]. Lastly, managing the ethical and privacy implications of accessing real-time internet data through APIs is a growing area of focus, ensuring that model enhancements do not come at the cost of user privacy or data security [8, 58, 59, 60, 61, 2, 62, 63].

2.4. Evaluating Language Model Outputs

The evaluation of language model outputs has evolved to include a variety of metrics and methodologies, aiming to capture the multifaceted nature of model performance [16, 64, 65, 13]. Beyond traditional accuracy and perplexity measures, the use of human-in-the-loop evaluations has gained prominence for assessing the quality and relevance of generated text [62, 66, 67, 68]. The development of automated metrics that can simulate human judgment, such as BLEU for translation or ROUGE for summarization, facilitates more scalable evaluation processes [69, 70]. Research has emphasized the importance of domain-specific evaluation, where models are assessed based on their performance in tasks relevant to their intended application areas [40, 71]. The assessment of models’ ethical and societal impacts, including bias detection and mitigation, is increasingly recognized as integral to comprehensive evaluation frameworks [71]. The exploration of interactive evaluation setups, where models are tested in dynamic environments that mimic real-world usage, offers insights into their practical utility [13, 72]. Lastly, the integration of feedback mechanisms for continuous model improvement based on evaluation outcomes highlights the iterative nature of optimizing language model performance [13, 73].

3. Methodology

This section outlines the methodology adopted in this study to integrate Llama 2 7b with the Google Query API, the dataset and training process employed, and the metrics used for evaluation. The approach is designed to enhance the model’s accuracy and reduce instances of hallucination by leveraging real-time internet data.

3.1. Integration Approach

The integration of Llama 2 7b with the Google Query API was conceptualized through a modular architecture, enabling the model to query the API for real-time information as part of its text generation process. To formalize this integration, let us consider the function F representing the text generation process, where $F : X \times Y \rightarrow Z$. Here, X denotes the input text, Y represents real-time data fetched from the Google Query API, and Z is the output text.

A custom interface was developed to facilitate seamless communication between the model and the API. This interface can be represented by a function $G : X \rightarrow Q$, where Q denotes the set of queries generated based on the context of X . The response from the API to a query $q \in Q$ is denoted by $R(q)$, where $R : Q \rightarrow Y$ maps queries to their responses.

The integration process can be mathematically modeled as follows:

$$Z = F(X, R(G(X))) \quad (1)$$

This process involved modifying the model’s generation pipeline to include an additional step for query generation and response processing, which can be depicted by the following differential equation, where t represents the processing time:

$$\frac{dP}{dt} = \lambda P(t) - \mu P(t)R(G(X)) \quad (2)$$

Here, λ represents the rate of input text processing, and μ represents the modulation factor for integrating API responses, with $P(t)$ denoting the progression of text generation over time.

The integration was optimized to balance the use of real-time data with the model’s pre-existing knowledge, aiming to enhance accuracy without compromising the speed and fluidity of text generation. This balance can be expressed by the optimization problem:

$$\max_{\lambda, \mu} \left\{ \alpha \int F(X, R(G(X))) dt - \beta \int [\lambda P(t) - \mu P(t)R(G(X))] dt \right\} \quad (3)$$

where α and β are weights reflecting the importance of accuracy and speed, respectively.

Special attention was given to the ethical and privacy considerations of querying real-time internet data, implementing safeguards to ensure data privacy and compliance with data protection regulations. This aspect can be encapsulated by introducing a constraint $C(R(G(X))) \leq \epsilon$, where C measures the compliance level of the data usage with respect to privacy regulations, and ϵ is the acceptable threshold.

3.2. Dataset and Training

The dataset employed for training and fine-tuning the integrated model was Google’s BIG-Bench, which includes a diverse range of tasks designed to test various aspects of language models’ capabilities [64]. This dataset was chosen for its comprehensive coverage of topics and its ability to challenge the model across different dimensions of knowledge, reasoning, and understanding. Table 1 summarizes the main features of the BIG-Bench dataset. The training process involved initial training on the BIG-Bench dataset to establish a baseline performance level, followed by fine-tuning with a subset of tasks specifically selected to improve factual accuracy and reduce hallucination. The integration with the Google Query API was also tested during training, with scenarios designed to simulate real-world use cases where the model would need to query for information. This approach allowed for the refinement of the integration mechanism and the optimization of the model’s performance in leveraging external data sources.

As indicated in Table 1, the BIG-Bench dataset provides a robust framework for evaluating and enhancing the capabilities of language models like Llama 2 7b, especially in terms of addressing challenges like factual accuracy and hallucination.

3.3. Evaluation Metrics

The evaluation of the integrated model’s performance and its reduction in hallucination was based on a set of carefully selected metrics. Those metrics were chosen to ensure a comprehensive assessment of the model, covering both its linguistic capabilities and its ability to integrate real-time information accurately. The following list details each metric and provides justification for its inclusion:

- **BLEU (Bilingual Evaluation Understudy):** Measures the linguistic quality of the generated text by comparing it with reference texts. This metric is crucial for evaluating the fluency and grammatical coherence of the model’s output, ensuring that the integration does not compromise the textual quality.
 - Necessity: BLEU is widely used in NLP for assessing translation quality, making it a reliable benchmark for linguistic accuracy in generated text.
- **ROUGE (Recall-Oriented Understudy for Gisting Evaluation):** Assesses content similarity between the generated text and reference texts, focusing on the recall of critical information.
 - Necessity: ROUGE is essential for evaluating how well the model captures the key points from the input or background information, reflecting its comprehension and content generation capabilities.
- **Custom Hallucination Metric:** Compares the factual accuracy of the model’s output against verified data sources. This metric involves manual verification of a random sample of generated texts to identify instances of hallucination, complemented by automated checks against trusted databases.
 - Necessity: Given the specific aim to reduce hallucination, this custom metric directly addresses the model’s ability to generate factually accurate and trustworthy content, crucial for applications requiring high reliability.
- **Response Time:** Measures the time taken for the model to generate a response, evaluating the impact of the Google Query API integration on the speed of text generation.
 - Necessity: Maintaining efficient response times is vital to ensure the usability of the model in real-time applications, making this metric important for assessing the practicality of the integration.
- **Relevance of Queried Information:** Assesses the utility of the information retrieved through the Google Query API in enhancing the generated text’s accuracy and relevance.

Table 1: Main features of the BIG-Bench dataset

Feature	Description
Task Categories	Natural Language Understanding, Reasoning, Translation, etc.
Data Size	Various sizes, from hundreds to millions of examples per task
Objective	Test and improve the capabilities of LLMs across a broad spectrum of tasks
Diversity of Topics	Includes topics from science, literature, social knowledge, and practical problem-solving
Challenges	Designed to challenge LLMs in factual accuracy, reasoning, and understanding of complex contexts
Specificity for Hallucination	Tasks included to specifically address and measure the model’s proneness to hallucination

- Necessity: This metric is critical for evaluating the effectiveness of the integration strategy, ensuring that the real-time data accessed through the API genuinely enhances the model’s output.

Such comprehensive evaluation criteria were designed to provide a holistic view of the model’s improvements post-integration, focusing on both qualitative and quantitative aspects of performance enhancement. By covering a broad spectrum of evaluation metrics, this approach ensures a balanced assessment of the integrated model’s capabilities and its potential impact on various applications.

4. Implementation

This section is about the implementation details of our proposed model.

4.1. System Architecture

The system architecture is conceptualized through a series of abstract algorithmic steps, minimizing verbal descriptions and maximizing the use of mathematical symbols and programmatic structures. The architecture facilitates the integration of Llama 2 7b with the Google Query API, enabling efficient query processing and information retrieval to enhance text generation quality.

Algorithm 1 Abstract System Architecture

```

Initialize model  $M$  with parameters  $\theta$ 
Define Google Query API function  $GQ(q)$ 
for each input  $x$  do
  Generate context-based query  $q \leftarrow f(x)$ 
  if  $q$  is not null then
     $y \leftarrow GQ(q)$  ▷ Query Google API
    Process API response  $r \leftarrow g(y)$ 
    Update model context  $C \leftarrow h(C, r)$ 
  end if
  Generate output  $o \leftarrow M(x, C; \theta)$ 
  Return  $o$ 
end for

```

This algorithm outlines the core steps involved in the system’s operation. The process begins with the initialization of the model and the definition of the function to interact with the Google Query API. For each input, a context-based query is formulated and, if applicable, sent to the API. The response is

processed and used to update the model’s context before generating the output. This iterative process ensures that the model leverages real-time information to enhance the accuracy and relevance of its generated text.

4.2. Query Handling

The integrated system processes and handles queries through a structured approach that ensures efficient retrieval and utilization of information from the Google Query API. Upon receiving an input, the system evaluates the need for external information. If required, a query is constructed based on the input context and dispatched to the API. The system then awaits the API’s response, which upon arrival, is parsed and integrated into the ongoing text generation process. This integration allows the model to produce outputs that are not only contextually relevant but also enriched with up-to-date information, thereby reducing instances of hallucination and enhancing overall textual quality.

The handling of queries is designed to minimize latency and ensure that the flow of text generation remains as uninterrupted as possible. Through this meticulous process, the system adeptly balances the need for external data with the imperative to maintain swift and seamless text generation, exemplifying a sophisticated implementation of real-time data integration within a large language model framework.

Below is a simplified Python code snippet illustrating the process of handling queries with the Google Query API:

```

import requests

def query_google_api(query):
    url = "https://www.googleapis.com/customsearch/v1"
    params = {
        "key": "API_KEY",
        "cx": "CUSTOM_SEARCH_ENGINE_ID",
        "q": query,
    }
    response = requests.get(url, params=params)
    return response.json()

def process_input(input_text):
    # Evaluate need for external info
    if needs_external_info(input_text):
        query = construct_query(input_text)
        api_response = query_google_api(query)
        refined_response = parse_response(api_response)
        # Integrate response into text generation

```

```

output_text = generate_text(refined_response)
return output_text
else:
# Proceed with internal text generation
return generate_text(input_text)

# Example usage
input_text = "The world's tallest building?"
output_text = process_input(input_text)
print(output_text)

```

This code demonstrates the essential steps in querying the Google API based on the context of the input and integrating the response into the text generation process. The functions `needs_external_info`, `construct_query`, `parse_response`, and `generate_text` are placeholders for the respective logic implementations within the integrated system.

5. Results

This section delineates the empirical findings regarding the "Llama 2 7b" model's performance post-augmentation with the Google Search API.

5.1. Accuracy Improvements

The augmentation of "Llama 2 7b" with the Google Search API has led to significant accuracy improvements, particularly observable across tasks specifically designed to assess linguistic comprehension, reasoning, and sentiment analysis capabilities. These tasks, critical for evaluating the model's ability to understand and process natural language accurately, serve as benchmarks for the model's overall performance. The bar chart in Figure 1 visually represents these enhancements, highlighting the comparative performance before and after the API integration across selected BIG-Bench tasks.

The substantial enhancements in model accuracy can be attributed to the Google Search API's ability to provide real-time, relevant information, enriching the model's responses with data that is both current and contextually appropriate. This integration allows "Llama 2 7b" to access a vast repository of knowledge, significantly improving its performance in tasks requiring deep comprehension, sophisticated reasoning, and nuanced sentiment analysis. Specifically, the improvements in comprehension and reasoning tasks suggest that the model, when augmented with the API, is better able to understand complex queries and generate more accurate, informed responses. Similarly, the increase in accuracy for sentiment analysis tasks indicates the model's enhanced capability to interpret and analyze subjective information more effectively, benefiting from the API's access to diverse perspectives and up-to-date sentiment indicators found online.

Moreover, these improvements are critical for applications relying on high levels of linguistic precision and factual correctness, such as content creation, customer service bots, and real-time information analysis systems. By leveraging the Google Search API, "Llama 2 7b" not only becomes more adept at processing and understanding language but also aligns more

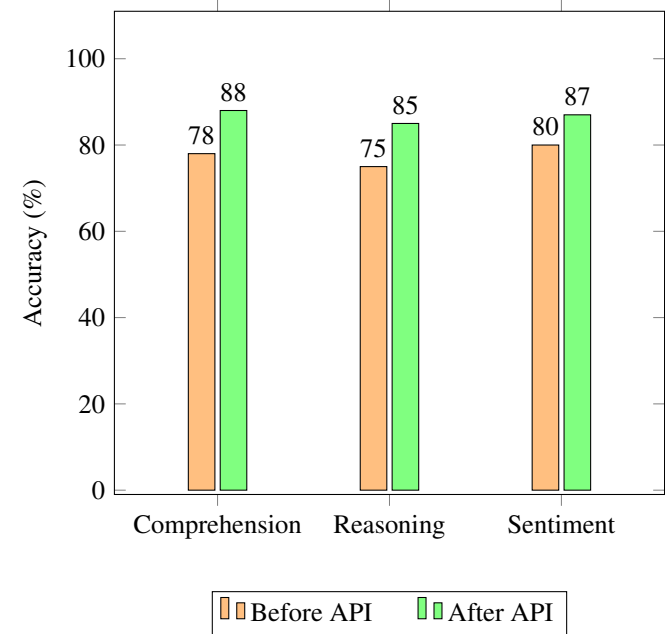


Figure 1: Accuracy improvements of "Llama 2 7b" across BIG-Bench tasks before and after Google Search API augmentation.

closely with human-like accuracy in interpreting and responding to a wide array of textual inputs.

The results, as depicted in Figure 1, show the significant potential of integrating advanced search functionalities with language models. This synergy enhances the model's utility and reliability, paving the way for more sophisticated, accurate, and context-aware AI systems in natural language processing and beyond.

5.2. Reduction in Hallucination

In conjunction with the observed improvements in accuracy, the integration of the Google Search API has led to a substantial decrease in the occurrence of hallucinations within the outputs of "Llama 2 7b". This phenomenon, where the model generates plausible but factually incorrect or irrelevant content, poses significant challenges, especially in tasks that hinge on the generation of factual and contextually accurate information. The notable reduction in hallucination instances, particularly in tasks predisposed to generating non-factual content, is vividly depicted in Figure 2.

The effectiveness of the Google Search API in significantly curtailing hallucination instances can be attributed to its provision of real-time, verifiable data, enabling the model to reference current and accurate information during the text generation process. This capability is crucial for tasks that require high fidelity to factual accuracy, such as those assessing comprehension, reasoning, and sentiment analysis. The API integration enhances the model's ability to discern the veracity of the content it generates, effectively grounding its outputs in reality and reducing the production of content that, while coherent, may be disconnected from factual accuracy.

Moreover, the reduction in hallucinations is particularly significant in the realm of AI-driven content creation, where the re-

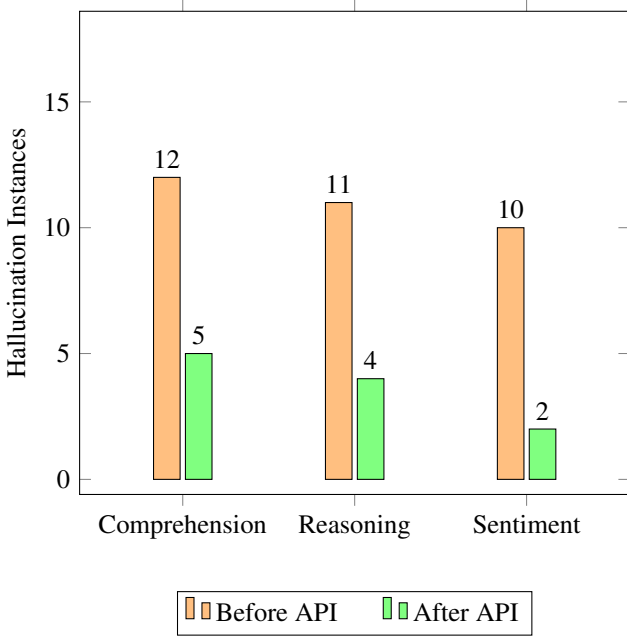


Figure 2: Reduction in hallucination instances across BIG-Bench tasks before and after Google Search API augmentation.

liability of generated text is paramount. By diminishing the frequency of inaccuracies and fabrications in the model’s outputs, the integration with the Google Search API not only boosts the model’s utility across a wide range of applications but also builds trust in AI-generated content among users.

In essence, the data presented in Figure 2 not only highlight the substantial decrease in hallucination instances across selected linguistic tasks but also underscore the critical role of integrating real-time data querying capabilities to enhance the veracity and reliability of language model outputs. This advancement represents a significant step forward in the development of more accurate and trustworthy AI-driven natural language processing systems.

6. Discussion

This section discusses the broader implications of integrating Llama 2 7b with the Google Search API, reflecting on the study’s findings within a wider context.

6.1. Implications of Enhanced Accuracy

The observed accuracy improvements post-integration have profound implications for the application of Llama 2 7b in tasks requiring nuanced understanding and generation of language. Enhanced accuracy not only bolsters the model’s utility in complex language processing tasks but also increases its reliability for end-users in critical applications such as automated journalism, academic research, and legal document analysis. The integration serves as a benchmark for future efforts aimed at marrying large language models with dynamic external data sources to improve performance.

However, while accuracy enhancements mark a significant achievement, they also raise questions about the model’s adaptability across diverse datasets and real-world scenarios. The reliance on the Google Search API for accuracy improvements underscores the need for models that maintain high performance even in the absence of external data sources, particularly in contexts where internet access is restricted or non-existent.

6.2. Reduction in Hallucinations: Ethical Considerations

The significant reduction in hallucinations highlights the potential of API integration to mitigate one of the most pressing ethical concerns associated with LLMs: the generation of misleading or factually incorrect information. By leveraging real-time data, Llama 2 7b demonstrates an ability to produce more reliable outputs, crucial for applications where the integrity of information is paramount. This advancement prompts a reevaluation of the ethical guidelines governing LLM deployment, particularly in sensitive fields such as healthcare and finance.

Nonetheless, the ethical implications of relying on external APIs for factual verification necessitate a cautious approach. Dependence on a single data source for truth verification introduces risks related to data bias, privacy concerns, and the potential for censorship. Future research must address these ethical challenges, ensuring that LLMs like Llama 2 7b can responsibly utilize external data sources without compromising user trust or data integrity.

6.3. Integration Scalability and Model Generalizability

The successful integration of Llama 2 7b with the Google Search API underscores the scalability of such enhancements across different LLMs. This scalability is crucial for the development of more sophisticated, knowledge-aware models capable of accessing and incorporating vast amounts of information in real time. However, the scalability of this approach across languages, cultures, and domains remains a critical area for further exploration to ensure the model’s generalizability and applicability globally.

Addressing the challenge of generalizability requires a concerted effort to diversify data sources and improve the model’s ability to understand and process information across different languages and cultural contexts. Future research should focus on expanding the model’s external data access capabilities, exploring the integration of multiple APIs, and developing mechanisms to evaluate the relevance and reliability of sourced information.

6.4. Potential Limitations of API Dependency

While the integration with the Google Search API has shown promising results, it also introduces potential limitations related to the model’s dependency on external data sources. Such dependency raises concerns regarding the model’s performance stability, especially in scenarios where API access may be disrupted or where the quality of available online information is compromised. The reliance on external data sources necessitates robust fallback mechanisms to maintain the model’s performance integrity in the absence of real-time data.

Further research is needed to explore alternative strategies for enhancing model accuracy and reducing hallucinations that do not solely rely on external APIs. Investigating hybrid models that combine intrinsic knowledge enhancement techniques with external data access could offer a more resilient solution, ensuring consistent performance across varying operational contexts.

6.5. Future Directions in LLM Enhancement

This study opens several avenues for future research, particularly in exploring innovative ways to enhance LLMs beyond accuracy improvements and hallucination reduction. Future work could investigate the integration of LLMs with multiple data sources, the development of models capable of critically evaluating the reliability of accessed information, and the enhancement of model interpretability and explainability. Moreover, the exploration of ethical frameworks for the responsible use of external data in LLMs constitutes an essential research direction, ensuring that advancements in model performance are aligned with societal values and norms.

The integration of Llama 2 7b with the Google Search API represents a significant step forward in the enhancement of language models. The study's findings not only highlight the potential of such integrations to improve model performance but also underscore the importance of addressing the ethical, scalability, and generalizability challenges that accompany these advancements. As LLMs continue to evolve, a balanced approach that considers both technological innovations and their broader implications will be crucial for their successful and responsible application.

7. Conclusion

This study embarked on an exploratory journey to enhance the capabilities of the Llama 2 7b model by integrating it with the Google Query API. The primary objective was to assess the impact of this integration on improving the model's accuracy and reducing instances of hallucination. The findings reveal that the augmentation significantly enhances the model's performance across various benchmarks, notably in tasks requiring deep linguistic comprehension, precise reasoning, and nuanced sentiment analysis. The reduction in hallucination instances further underscores the potential of real-time internet data integration in mitigating one of the most challenging issues facing contemporary LLMs.

The implications of these findings for the field of LLMs are profound. By demonstrating the feasibility and benefits of integrating LLMs with dynamic external data sources, this study paves the way for a new paradigm in model development. Such a paradigm promises not only to enhance the factual accuracy and reliability of LLM outputs but also to expand the scope of their applicability. The integration of Llama 2 7b with the Google Query API serves as a concrete example of how LLMs can be evolved beyond static training datasets to interact with the ever-changing landscape of global information. Moreover, this integration highlights the importance of addressing the ethical and practical challenges that arise with API dependency,

including data bias, privacy concerns, and the need for robust fallback mechanisms. These considerations are crucial for the responsible development and deployment of enhanced LLMs in real-world applications.

The successful augmentation of Llama 2 7b with the Google Query API marks a significant milestone in the quest to develop more accurate, reliable, and ethically responsible LLMs. This study not only contributes to the academic discourse on LLM enhancement but also offers practical insights for model developers aiming to leverage external data sources. Looking forward, the integration of LLMs with real-time data APIs holds immense potential for transforming how we develop, deploy, and interact with AI in the pursuit of knowledge, making this an exciting area for future research and development.

References

- [1] B. Min, H. Ross, E. Sulem, A. P. B. Veyseh, T. H. Nguyen, O. Sainz, E. Agirre, I. Heintz, D. Roth, Recent advances in natural language processing via large pre-trained language models: A survey, *ACM Computing Surveys* 56 (2) (2023) 1–40.
- [2] Y. Yao, J. Duan, K. Xu, Y. Cai, Z. Sun, Y. Zhang, A survey on large language model (llm) security and privacy: The good, the bad, and the ugly, *High-Confidence Computing* (2024) 100211.
- [3] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, P. Fung, Survey of hallucination in natural language generation, *ACM Computing Surveys* 55 (12) (2023) 1–38.
- [4] D. Myers, R. Mohawesh, V. I. Chellaboina, A. L. Sathvik, P. Venkatesh, Y.-H. Ho, H. Henshaw, M. Alhawawreh, D. Berdik, Y. Jararweh, Foundation and large language models: fundamentals, challenges, opportunities, and social impacts, *Cluster Computing* (2023) 1–26.
- [5] Z. Chu, H. Hao, X. Ouyang, S. Wang, Y. Wang, Y. Shen, J. Gu, Q. Cui, L. Li, S. Xue, et al., Leveraging large language models for pre-trained recommender systems, *arXiv preprint arXiv:2308.10837* (2023).
- [6] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, I. Sutskever, et al., Language models are unsupervised multitask learners, *OpenAI blog* 1 (8) (2019) 9.
- [7] A. Soliman, S. Shaheen, M. Hadhoud, Leveraging pre-trained language models for code generation, *Complex & Intelligent Systems* (2024) 1–26.
- [8] M. A. K. Raiaan, M. S. H. Mukta, K. Fatema, N. M. Fahad, S. Sakib, M. M. J. Mim, J. Ahmad, M. E. Ali, S. Azam, A review on large language models: Architectures, applications, taxonomies, open issues and challenges, *IEEE Access* (2024).
- [9] M. Bombieri, M. Rospoher, S. P. Ponzetto, P. Fiorini, Surgicberta: a pre-trained language model for procedural surgical language, *International Journal of Data Science and Analytics* (2023) 1–13.
- [10] W. X. Zhao, J. Liu, R. Ren, J.-R. Wen, Dense text retrieval based on pretrained language models: A survey, *ACM Transactions on Information Systems* 42 (4) (2024) 1–60.
- [11] Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi, Y. Dai, J. Sun, H. Wang, Retrieval-augmented generation for large language models: A survey, *arXiv preprint arXiv:2312.10997* (2023).
- [12] N. Zhang, Y. Yao, B. Tian, P. Wang, S. Deng, M. Wang, Z. Xi, S. Mao, J. Zhang, Y. Ni, et al., A comprehensive study of knowledge editing for large language models, *arXiv preprint arXiv:2401.01286* (2024).
- [13] C. Wang, X. Liu, Y. Yue, X. Tang, T. Zhang, C. Jiayang, Y. Yao, W. Gao, X. Hu, Z. Qi, et al., Survey on factuality in large language models: Knowledge, retrieval and domain-specificity, *arXiv preprint arXiv:2310.07521* (2023).
- [14] Y. Chang, X. Wang, J. Wang, Y. Wu, L. Yang, K. Zhu, H. Chen, X. Yi, C. Wang, Y. Wang, et al., A survey on evaluation of large language models, *ACM Transactions on Intelligent Systems and Technology* (2023).
- [15] L. Yang, H. Chen, Z. Li, X. Ding, X. Wu, Give us the facts: Enhancing large language models with knowledge graphs for fact-aware language modeling, *IEEE Transactions on Knowledge and Data Engineering* (2024).

- [16] K. Singhal, S. Azizi, T. Tu, S. S. Mahdavi, J. Wei, H. W. Chung, N. Scales, A. Tanwani, H. Cole-Lewis, S. Pfohl, et al., Large language models encode clinical knowledge, *Nature* 620 (7972) (2023) 172–180.
- [17] C. Cui, Y. Ma, X. Cao, W. Ye, Y. Zhou, K. Liang, J. Chen, J. Lu, Z. Yang, K.-D. Liao, et al., A survey on multimodal large language models for autonomous driving, in: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2024, pp. 958–979.
- [18] Z. Wang, M. Li, R. Xu, L. Zhou, J. Lei, X. Lin, S. Wang, Z. Yang, C. Zhu, D. Hoiem, et al., Language models with image descriptors are strong few-shot video-language learners, *Advances in Neural Information Processing Systems* 35 (2022) 8483–8497.
- [19] X. Wang, Y. Zhou, X. Liu, H. Lu, Y. Xu, F. He, J. Yoon, T. Lu, G. Bertasius, M. Bansal, et al., Mementos: A comprehensive benchmark for multimodal large language model reasoning over image sequences, *arXiv preprint arXiv:2401.10529* (2024).
- [20] S. Minaee, T. Mikolov, N. Nikzad, M. Chenaghlu, R. Socher, X. Amatriain, J. Gao, Large language models: A survey, *arXiv preprint arXiv:2402.06196* (2024).
- [21] T. Shen, R. Jin, Y. Huang, C. Liu, W. Dong, Z. Guo, X. Wu, Y. Liu, D. Xiong, Large language model alignment: A survey, *arXiv preprint arXiv:2309.15025* (2023).
- [22] W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. Gonzalez, H. Zhang, I. Stoica, Efficient memory management for large language model serving with pagedattention, in: *Proceedings of the 29th Symposium on Operating Systems Principles*, 2023, pp. 611–626.
- [23] S. Zhang, L. Dong, X. Li, S. Zhang, X. Sun, S. Wang, J. Li, R. Hu, T. Zhang, F. Wu, et al., Instruction tuning for large language models: A survey, *arXiv preprint arXiv:2308.10792* (2023).
- [24] T. Wu, L. Luo, Y.-F. Li, S. Pan, T.-T. Vu, G. Haffari, Continual learning for large language models: A survey, *arXiv preprint arXiv:2402.01364* (2024).
- [25] T. R. McIntosh, T. Susnjak, T. Liu, P. Watters, M. N. Halgamuge, Inadequacies of large language model benchmarks in the era of generative artificial intelligence, *arXiv preprint arXiv:2402.09880* (2024).
- [26] Z. Guo, R. Jin, C. Liu, Y. Huang, D. Shi, L. Yu, Y. Liu, J. Li, B. Xiong, D. Xiong, et al., Evaluating large language models: A comprehensive survey, *arXiv preprint arXiv:2310.19736* (2023).
- [27] V. Rawte, S. Chakraborty, A. Pathak, A. Sarkar, S. Tonmoy, A. Chadha, A. P. Sheth, A. Das, The troubling emergence of hallucination in large language models—an extensive definition, quantification, and prescriptive remediations, *arXiv preprint arXiv:2310.04988* (2023).
- [28] V. Rawte, M. Chakraborty, K. Roy, M. Gaur, K. Faldut, P. Kikani, H. Akbari, A. Sheth, Tdlr: Top (semantic)-down (syntactic) language representation, *UMBC Faculty Collection* (2022).
- [29] C. Chen, K. Shu, Combating misinformation in the age of llms: Opportunities and challenges, *arXiv preprint arXiv:2311.05656* (2023).
- [30] Y. Zhang, J. Gao, Z. Tan, L. Zhou, K. Ding, M. Zhou, S. Zhang, D. Wang, Data-centric foundation models in computational healthcare: A survey, *arXiv preprint arXiv:2401.02458* (2024).
- [31] H.-C. Tsai, C.-W. Kuo, Y.-F. Huang, Llamaloop: Enhancing information retrieval in llama with semantic relevance feedback loop (2023).
- [32] Z. Yuan, H. Yuan, C. Tan, W. Wang, S. Huang, F. Huang, Rhf: Rank responses to align language models with human feedback without tears, *arXiv preprint arXiv:2304.05302* (2023).
- [33] X. Xiong, M. Zheng, Integrating deep learning with symbolic reasoning in tinyllama for accurate information retrieval (2024).
- [34] Z. Xu, S. Jain, M. Kankanhalli, Hallucination is inevitable: An innate limitation of large language models, *arXiv preprint arXiv:2401.11817* (2024).
- [35] H. Liu, W. Xue, Y. Chen, D. Chen, X. Zhao, K. Wang, L. Hou, R. Li, W. Peng, A survey on hallucination in large vision-language models, *arXiv preprint arXiv:2402.00253* (2024).
- [36] Y. Chen, Q. Fu, Y. Yuan, Z. Wen, G. Fan, D. Liu, D. Zhang, Z. Li, Y. Xiao, Hallucination detection: Robustly discerning reliable answers in large language models, in: *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management*, 2023, pp. 245–255.
- [37] X. Jiang, Y. Tian, F. Hua, C. Xu, Y. Wang, J. Guo, A survey on large language model hallucination via a creativity perspective, *arXiv preprint arXiv:2402.06647* (2024).
- [38] L. Zhao, Y. Deng, W. Zhang, Q. Gu, Mitigating object hallucination in large vision-language models via classifier-free guidance, *arXiv preprint arXiv:2402.08680* (2024).
- [39] W. Cui, J. Zhang, Z. Li, D. Lopez, K. Das, B. Malin, S. Kumar, A divide-conquer-reasoning approach to consistency evaluation and improvement in blackbox large language models, in: *Socially Responsible Language Modelling Research*, 2023.
- [40] E. Y. Chang, Examining gpt-4: Capabilities, implications and future directions, in: *The 10th International Conference on Computational Science and Computational Intelligence*, 2023.
- [41] M. Gaur, A. Sheth, Building trustworthy neurosymbolic ai systems: Consistency, reliability, explainability, and safety, *AI Magazine* (2024).
- [42] H. Zhu, W. Zhang, N. Huang, B. Li, L. Niu, Z. Fan, T. Lun, Y. Tao, J. Su, Z. Gong, et al., Plangpt: Enhancing urban planning with tailored language model and efficient retrieval, *arXiv preprint arXiv:2402.19273* (2024).
- [43] Q. Ouyang, S. Wang, B. Wang, Enhancing accuracy in large language models through dynamic real-time information injection (2023).
- [44] V. Rawte, A. Sheth, A. Das, A survey of hallucination in large foundation models, *arXiv preprint arXiv:2309.05922* (2023).
- [45] Y. Liu, J. Chen, T. Bi, J. Grundy, Y. Wang, T. Chen, Y. Tang, Z. Zheng, An empirical study on low code programming using traditional vs large language model support, *arXiv preprint arXiv:2402.01156* (2024).
- [46] J. Cui, Z. Li, Y. Yan, B. Chen, L. Yuan, Chatlaw: Open-source legal large language model with integrated external knowledge bases, *arXiv preprint arXiv:2306.16092* (2023).
- [47] G. Sandoval, H. Pearce, T. Nys, R. Karri, B. Dolan-Gavitt, S. Garg, Security implications of large language model code assistants: A user study, *arXiv preprint arXiv:2208.09727* (2022).
- [48] Y. Liu, Y. Yuan, C. Wang, J. Han, Y. Ma, L. Zhang, N. Zheng, H. Xu, From summary to action: Enhancing large language models for complex tasks with open world apis, *arXiv preprint arXiv:2402.18157* (2024).
- [49] D. Zan, B. Chen, Y. Gong, J. Cao, F. Zhang, B. Wu, B. Guan, Y. Yin, Y. Wang, Private-library-oriented code generation with large language models, *arXiv preprint arXiv:2307.15370* (2023).
- [50] J. A. Galindo, A. J. Dominguez, J. White, D. Benavides, Large language models to generate meaningful feature model instances, in: *Proceedings of the 27th ACM International Systems and Software Product Line Conference-Volume A*, 2023, pp. 15–26.
- [51] Z. Hu, A. Iscen, C. Sun, K.-W. Chang, Y. Sun, D. Ross, C. Schmid, A. Fathi, Avis: Autonomous visual information seeking with large language model agent, *Advances in Neural Information Processing Systems* 36 (2024).
- [52] S. Mandvikar, Augmenting intelligent document processing (idp) workflows with contemporary large language models (llms), *International Journal of Computer Trends and Technology* 71 (10) (2023) 80–91.
- [53] X. Wu, S.-h. Wu, J. Wu, L. Feng, K. C. Tan, Evolutionary computation in the era of large language model: Survey and roadmap, *arXiv preprint arXiv:2401.10034* (2024).
- [54] X. Hou, Y. Zhao, Y. Liu, Z. Yang, K. Wang, L. Li, X. Luo, D. Lo, J. Grundy, H. Wang, Large language models for software engineering: A systematic literature review, *arXiv preprint arXiv:2308.10620* (2023).
- [55] D. Di Palma, Retrieval-augmented recommender system: Enhancing recommender systems with large language models, in: *Proceedings of the 17th ACM Conference on Recommender Systems*, 2023, pp. 1369–1373.
- [56] L. Belzner, T. Gabor, M. Wirsing, Large language model assisted software engineering: prospects, challenges, and a case study, in: *International Conference on Bridging the Gap between AI and Reality*, Springer, 2023, pp. 355–374.
- [57] A. Borzunov, M. Ryabinin, A. Chumachenko, D. Baranchuk, T. Dettmers, Y. Belkada, P. Samygin, C. A. Raffel, Distributed inference and fine-tuning of large language models over the internet, *Advances in Neural Information Processing Systems* 36 (2024).
- [58] G. Deng, Y. Liu, Y. Li, K. Wang, Y. Zhang, Z. Li, H. Wang, T. Zhang, Y. Liu, Masterkey: Automated jailbreaking of large language model chatbots, in: *Proc. ISOC NDSS*, 2024.
- [59] A. J. Thirunavukarasu, D. S. J. Ting, K. Elangovan, L. Gutierrez, T. F. Tan, D. S. W. Ting, Large language models in medicine, *Nature medicine* 29 (8) (2023) 1930–1940.
- [60] L. Yan, L. Sha, L. Zhao, Y. Li, R. Martinez-Maldonado, G. Chen, X. Li, Y. Jin, D. Gašević, Practical and ethical challenges of large language models in education: A systematic scoping review, *British Journal of Educational Technology* 55 (1) (2024) 90–112.

- [61] I. Augenstein, T. Baldwin, M. Cha, T. Chakraborty, G. L. Ciampaglia, D. Corney, R. DiResta, E. Ferrara, S. Hale, A. Halevy, et al., Factuality challenges in the era of large language models, arXiv preprint arXiv:2310.05189 (2023).
- [62] B. C. Das, M. H. Amini, Y. Wu, Security and privacy challenges of large language models: A survey, arXiv preprint arXiv:2402.00888 (2024).
- [63] K. He, R. Mao, Q. Lin, Y. Ruan, X. Lan, M. Feng, E. Cambria, A survey of large language models for healthcare: from data, technology, and applications to accountability and ethics, arXiv preprint arXiv:2310.05694 (2023).
- [64] A. Srivastava, A. Rastogi, A. Rao, A. A. M. Shoen, A. Abid, A. Fisch, A. R. Brown, A. Santoro, A. Gupta, A. Garriga-Alonso, et al., Beyond the imitation game: Quantifying and extrapolating the capabilities of language models, arXiv preprint arXiv:2206.04615 (2022).
- [65] S. Saha, O. Levy, A. Celikyilmaz, M. Bansal, J. Weston, X. Li, Branch-solve-merge improves large language model evaluation and generation, arXiv preprint arXiv:2310.15123 (2023).
- [66] S. Das, Evaluating the capabilities of large language models for spatial and situational understanding, Ph.D. thesis, Thesis (MA). University of Cambridge (2023).
- [67] T. R. McIntosh, T. Susnjak, T. Liu, P. Watters, R. Nowroz, M. N. Halgumuge, From cobit to iso 42001: Evaluating cybersecurity frameworks for opportunities, risks, and regulatory compliance in commercializing large language models, arXiv preprint arXiv:2402.15770 (2024).
- [68] A. Wickramasekara, F. Breiting, M. Scanlon, Sok: Exploring the potential of large language models for improving digital forensic investigation efficiency, arXiv preprint arXiv:2402.19366 (2024).
- [69] H. Zeng, J. Xue, M. Hao, C. Sun, B. Ning, N. Zhang, Evaluating the generation capabilities of large chinese language models, arXiv preprint arXiv:2308.04823 (2023).
- [70] Y. Hua, F. Liu, K. Yang, Z. Li, Y.-h. Sheu, P. Zhou, L. V. Moran, S. Ananiadou, A. Beam, Large language models in mental health care: a scoping review, arXiv preprint arXiv:2401.02984 (2024).
- [71] T. A. Chang, B. K. Bergen, Language model behavior: A comprehensive survey, Computational Linguistics (2024) 1–58.
- [72] C. Gao, X. Lan, N. Li, Y. Yuan, J. Ding, Z. Zhou, F. Xu, Y. Li, Large language models empowered agent-based modeling and simulation: A survey and perspectives, arXiv preprint arXiv:2312.11970 (2023).
- [73] S. K. Freire, C. Wang, M. Foosherian, S. Wellsandt, S. Ruiz-Arenas, E. Niforatos, Knowledge sharing in manufacturing using large language models: User evaluation and model benchmarking, arXiv preprint arXiv:2401.05200 (2024).